

NLP-Based Movie Genre Classification Using TF-IDF and Machine Learning

Mini NLP Project Report

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1. Title

NLP-Based Movie Genre Classification Using TF-IDF and Machine Learning

2. Problem Statement

Movies are categorized into different genres such as Drama, Comedy, Horror, Action, Thriller, Documentary, and Western. When working with a large collection of movies, manually identifying the appropriate genre for each movie can be time-consuming.

Movie descriptions contain useful information about the storyline, events, characters, and themes of a movie. Therefore, Natural Language Processing (NLP) can be used to analyze these descriptions and identify patterns associated with different movie genres.

This project develops an NLP-based machine learning system that automatically predicts the genre of a movie based on its textual description.

The system treats movie genre prediction as a multi-class text classification problem. The movie description is provided as input, and the trained model predicts one of the selected movie genres as the output.

The project demonstrates how NLP and machine learning can be combined to automate the classification of movie descriptions.

3. Objectives

The main objectives of this project are:

1. To develop an NLP-based system for automatic movie genre classification.
2. To use movie descriptions as textual input for classification.
3. To clean and preprocess the movie description data.
4. To convert textual data into numerical features using TF-IDF.
5. To implement Logistic Regression for movie genre classification.
6. To implement Multinomial Naive Bayes for comparison.
7. To evaluate both models using accuracy, precision, recall, F1-score, and confusion matrix.
8. To compare the performance of the two classification models.
9. To select the better-performing model.
10. To create a system that can predict the genre of a new movie description.

4. Dataset

4.1 Dataset

The dataset used in this project is a **Movies Genre Description dataset** containing information about movies and their genres.

The dataset contains movie titles, genre labels, movie descriptions, and date information. The movie description is the main textual information used for the NLP classification task.

The dataset is suitable for this project because the objective is to predict a movie's genre from its description.

4.2 Dataset Source

The dataset was obtained as a publicly available CSV dataset and used locally in Google Colab for the project.

The dataset file used in the project is:

Movies_Genre_Description.csv

4.3 Dataset Size

The original dataset contains:

- **108,414 movie records**
- **4 attributes**

The four attributes are:

<i>Attribute</i>	<i>Description</i>
<i>TITLE</i>	Name of the movie
<i>GENRE</i>	Genre/category of the movie
<i>DESCRIPTION</i>	Textual description or plot summary of the movie
<i>DATE</i>	Release date/year information

The original dataset contains **28 unique genre categories**.

For this mini-project, eight well-represented genres were selected to make the classification problem manageable while maintaining sufficient training examples for each class.

The selected genres are:

1. Action
2. Comedy
3. Documentary
4. Drama
5. Horror
6. Short
7. Thriller

8. Western

After genre selection and data cleaning, the final dataset used for the experiment contained **90,496 movie records**.

4.4 Input and Target

The project uses:

Input: Movie description (DESCRIPTION)

Target: Movie genre (GENRE)

Therefore, the classification process can be represented as:

Movie Description → Predicted Movie Genre

4.5 Data Preparation

The dataset was first inspected to understand its structure and identify possible data-quality issues.

The following preparation steps were performed:

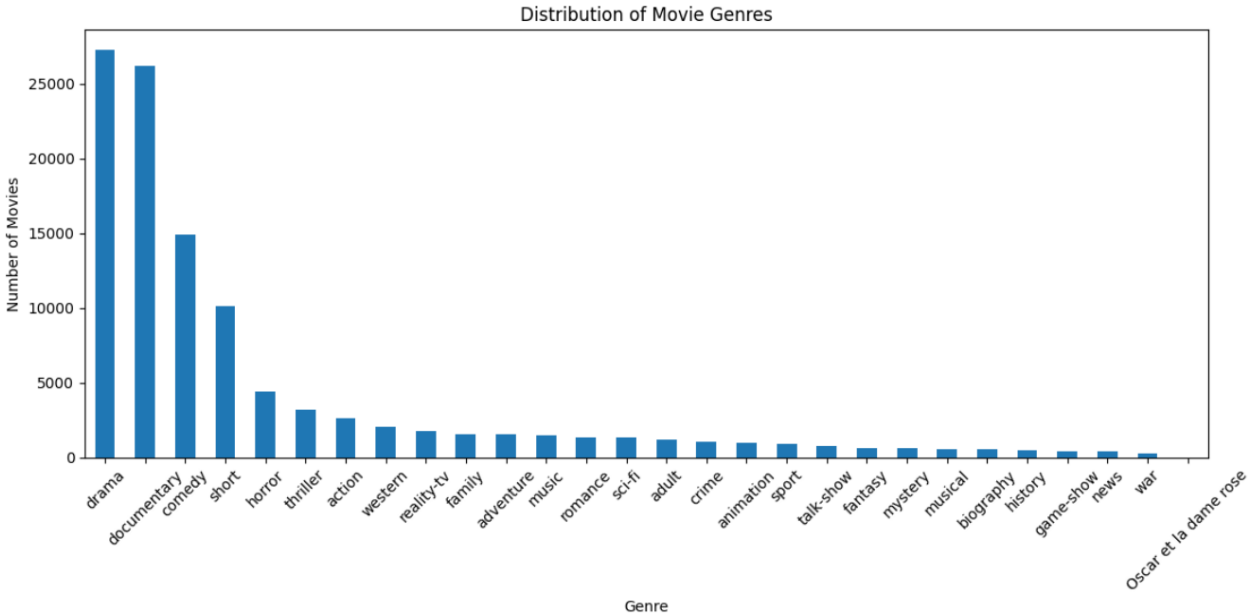
1. The CSV dataset was loaded into Python using Pandas.
2. The dataset dimensions and column names were examined.
3. Missing values were checked.
4. Duplicate records were identified.
5. Eight well-represented genres were selected.
6. Records outside the selected genres were removed.
7. Records with missing descriptions or genres were removed.
8. Duplicate movie descriptions were removed.
9. The description column was selected as the main textual input.
10. The cleaned descriptions were used for further NLP processing.

The final cleaned dataset was then divided into training and testing data.

Screenshot 1: Dataset Loading and Initial Exploration

	TITLE	GENRE	DESCRIPTION	DATE
0	Oscar et la dame rose	Oscar et la dame rose	Listening in to a conversation between his doc...	2009
1	Cupid	thriller	A brother and sister with a past incestuous re...	1997
2	Young, Wild and Wonderful	adult	As the bus empties the students for their fiel...	1980
3	The Secret Sin	drama	To help their unemployed father make ends meet...	1915
4	The Unrecovered	drama	The film's title refers not only to the un-rec...	2007

Screenshot 2: Genre Distribution



5. Methodology

The overall methodology followed in this project is:

Movie Description → Data Cleaning → Text Preprocessing → Train-Test Split → TF-IDF → Classification Model → Prediction → Evaluation

5.1 Data Loading

The dataset was uploaded to Google Colab using the file-upload functionality. Pandas was then used to load the CSV file into a DataFrame.

5.2 Data Exploration

The dataset was explored by checking its shape, column names, sample records, missing values, duplicate records, and genre distribution.

This step helped in understanding the structure and quality of the dataset before applying NLP techniques.

5.3 Genre Selection

The original dataset contained 28 different genre categories. Since some categories had relatively few examples, eight well-represented genres were selected for the mini-project.

The selected categories were:

- Action
- Comedy
- Documentary
- Drama
- Horror
- Short
- Thriller
- Western

This reduced the complexity of the classification task while still providing a meaningful multi-class NLP problem.

5.4 Text Preprocessing

The movie descriptions were preprocessed before feature extraction.

The following operations were performed:

- Conversion of text to lowercase
- Removal of special characters and unnecessary symbols

- Splitting text into individual words
- Removal of English stopwords
- Removal of empty descriptions after preprocessing

Preprocessing reduces unnecessary variation in the text and prepares the descriptions for machine learning.

Screenshot 3: Text Preprocessing

	DESCRIPTION	CLEANED_DESCRIPTION
1	A brother and sister with a past incestuous re...	brother sister past incestuous relationship cu...
3	To help their unemployed father make ends meet...	help unemployed father make ends meet edith tw...
4	The film's title refers not only to the un-rec...	films title refers unrecovered bodies ground z...
5	Quality Control consists of a series of 16mm s...	quality control consists series mm single take...
6	In tough economic times Max and Joey have all ...	tough economic times max joey run ideas discov...

5.5 Train-Test Split

The cleaned dataset was divided into training and testing sets.

The split used was:

Dataset Number of Records

<i>Training</i>	72,396
<i>Testing</i>	18,100
<i>Total</i>	90,496

An 80:20 train-test split was used, with stratification to maintain the distribution of genres in both sets.

5.6 TF-IDF Feature Extraction

Term Frequency-Inverse Document Frequency (TF-IDF) was used to convert the movie descriptions into numerical feature vectors.

TF-IDF assigns higher importance to words that are useful for distinguishing documents while reducing the importance of words that occur frequently across many documents.

The TF-IDF vectorizer used:

- Maximum features: 10,000
- Unigrams and bigrams
- Minimum document frequency: 2

The resulting training and testing data contained **10,000 TF-IDF features**.

The TF-IDF transformation can be represented as:

Preprocessed Text → TF-IDF Vector → Numerical Features

Screenshot 4: TF-IDF Feature Extraction

```
Training samples: 72396  
Testing samples: 18100
```

5.7 Classification

Two machine learning algorithms were implemented:

Logistic Regression

Logistic Regression was used as one of the main classification algorithms. It is a commonly used algorithm for text classification because it can work effectively with high-dimensional sparse feature representations such as TF-IDF.

Multinomial Naive Bayes

Multinomial Naive Bayes was used as the second classification algorithm. It is a simple and efficient probabilistic algorithm that is widely used for text classification tasks.

The two models were trained using the same TF-IDF features so that their performance could be fairly compared.

5.8 Prediction and Evaluation

After training, both models were used to predict genres for the test descriptions.

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The model with the better overall performance was selected as the final model.

6. Implementation

6.1 Programming Language

The project was implemented using **Python**.

The development and execution environment used was **Google Colab**.

6.2 Libraries Used

The main Python libraries used in the project were:

<i>Library</i>	<i>Purpose</i>
<i>Pandas</i>	Dataset loading and manipulation
<i>NumPy</i>	Numerical operations
<i>Matplotlib</i>	Data visualization
<i>Seaborn</i>	Confusion matrix visualization
<i>Scikit-learn</i>	TF-IDF, model training, and evaluation
<i>NLTK</i>	Stopword processing
<i>re</i>	Text cleaning using regular expressions

6.3 NLP Techniques

The main NLP techniques used were:

1. Lowercase conversion
2. Special-character removal
3. Stopword removal
4. TF-IDF feature extraction
5. Unigram and bigram representation

6.4 Machine Learning Algorithms

Two supervised machine learning algorithms were implemented:

1. **Logistic Regression**
2. **Multinomial Naive Bayes**

Both models were trained using the TF-IDF representation of the movie descriptions.

6.5 Important Implementation Details

The project used an 80:20 train-test split.

The training data was used to fit the TF-IDF vectorizer, while the test data was transformed using the already-fitted vectorizer. This prevents information from the test set from being used during feature learning.

The Logistic Regression model was configured with a maximum of 1,000 iterations to allow sufficient iterations for convergence.

The final system also includes a prediction function that accepts a new movie description, applies the same preprocessing and TF-IDF transformation, and returns the predicted genre.

7. Results

The two classification models were trained and evaluated on **18,100 test movie descriptions**.

7.1 Model Accuracy

The accuracy obtained by the two models is shown below:

<i>Model</i>	<i>Accuracy</i>
<i>Logistic Regression</i>	69.84%
<i>Multinomial Naive Bayes</i>	63.55%

Logistic Regression achieved the highest accuracy of **69.84%**, while Multinomial Naive Bayes achieved **63.55%**.

Therefore, Logistic Regression was selected as the final model.

Screenshot 5: Logistic Regression Results

```
Logistic Regression Accuracy: 0.6984
Logistic Regression Accuracy: 69.84%
```

```
Logistic Regression Classification Report:
              precision    recall  f1-score   support

   action          0.69      0.34      0.46       525
    comedy          0.65      0.65      0.65      2971
documentary          0.79      0.88      0.83      5231
    drama          0.65      0.79      0.71      5422
    horror          0.76      0.63      0.69       881
     short          0.59      0.35      0.44      2023
    thriller          0.50      0.15      0.24       635
    western          0.92      0.75      0.83       412

 accuracy                   0.70      18100
 macro avg          0.69      0.57      0.61      18100
weighted avg          0.69      0.70      0.68      18100
```

Screenshot 6: Naive Bayes Results

```
Naive Bayes Accuracy: 0.6355
Naive Bayes Accuracy: 63.55%
```

```
Naive Bayes Classification Report:
              precision    recall  f1-score   support

   action          0.73      0.11      0.19       525
    comedy          0.66      0.49      0.56      2971
documentary          0.71      0.88      0.79      5231
    drama          0.55      0.82      0.66      5422
    horror          0.81      0.43      0.56       881
     short          0.68      0.14      0.23      2023
    thriller          0.52      0.02      0.04       635
    western          0.98      0.65      0.78       412

 accuracy                   0.64      18100
 macro avg          0.70      0.44      0.48      18100
weighted avg          0.65      0.64      0.59      18100
```

7.2 Model Comparison

Logistic Regression performed better than Multinomial Naive Bayes.

The difference in accuracy was approximately:

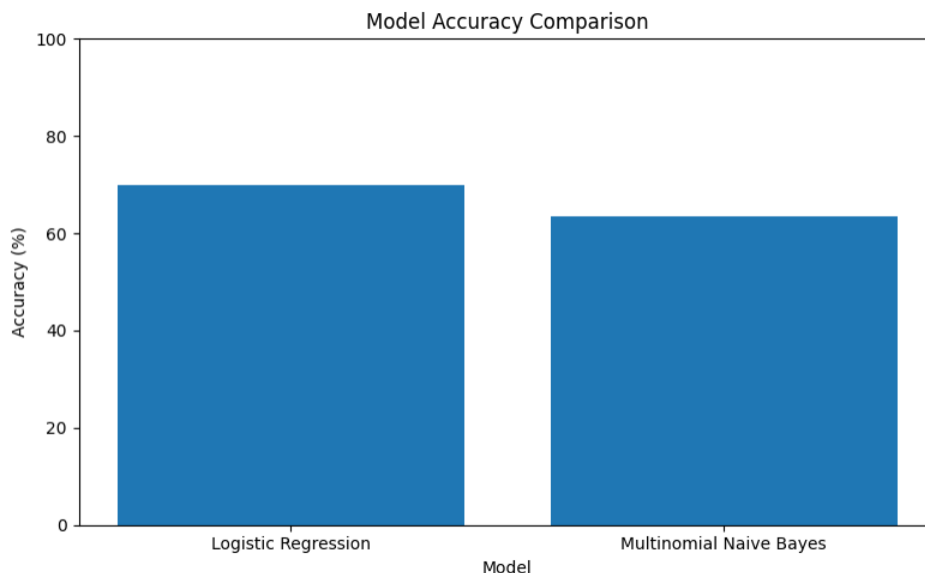
$$69.84\% - 63.55\% = 6.29 \text{ percentage points}$$

This indicates that Logistic Regression was better suited to the selected TF-IDF representation for this movie genre classification task.

Screenshot 7: Model Comparison

	Model	Accuracy	Accuracy (%)
0	Logistic Regression	0.698398	69.839779
1	Multinomial Naive Bayes	0.635470	63.546961

Screenshot 8: Accuracy Comparison Graph



7.3 Logistic Regression Classification Results

The classification report for Logistic Regression produced the following results:

<i>Genre</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
Action	0.69	0.34	0.46

<i>Comedy</i>	0.65	0.65	0.65
<i>Documentary</i>	0.79	0.88	0.83
<i>Drama</i>	0.65	0.79	0.71
<i>Horror</i>	0.76	0.63	0.69
<i>Short</i>	0.59	0.35	0.44
<i>Thriller</i>	0.50	0.15	0.24
<i>Western</i>	0.92	0.75	0.83

The overall accuracy of the Logistic Regression model was **69.84%**.

The model performed particularly well for **Documentary and Western**, while **Thriller, Short, and Action** were more difficult to classify.

7.4 Multinomial Naive Bayes Classification Results

The Multinomial Naive Bayes model achieved an overall accuracy of **63.55%**.

Its performance was lower than Logistic Regression for the selected classification task.

7.5 Confusion Matrix

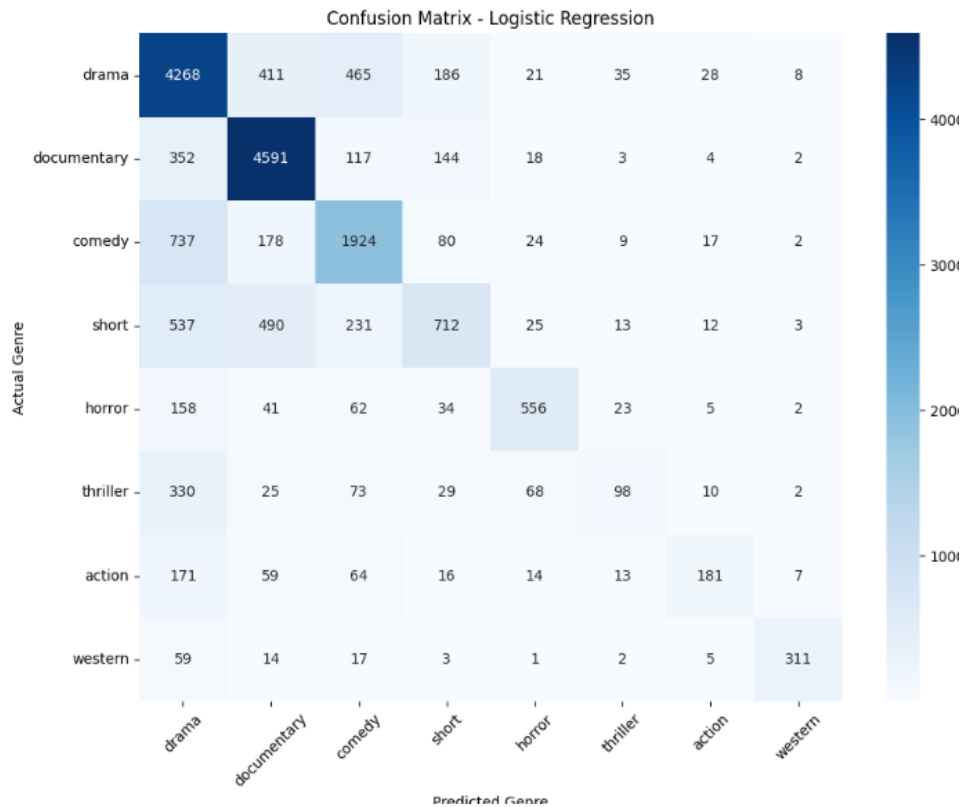
A confusion matrix was generated for the best-performing Logistic Regression model.

The confusion matrix shows the relationship between the actual movie genres and the genres predicted by the model.

It helps identify which genres are correctly classified and which genres are frequently confused with one another.

For example, some confusion can occur between genres such as Action, Horror, and Thriller because their movie descriptions may contain similar words and themes.

Screenshot 9: Confusion Matrix



7.6 Sample Predictions

The final model was also tested using new movie descriptions.

Example:

Movie Description	Predicted Genre
<i>A detective investigates a series of mysterious murders in a small town.</i>	Model Prediction
<i>A group of friends spend the weekend at an abandoned house where strange events begin to happen.</i>	Model Prediction
<i>Two people from different backgrounds meet and gradually fall in love.</i>	Model Prediction
<i>A family goes on an exciting journey across the country and faces unexpected adventures.</i>	Model Prediction

The final notebook also provides an interactive input system where a user can enter a new movie description and receive a predicted genre.

Screenshot 10: Final Prediction

```
Description:
A detective follows clues to solve a mysterious murder involving several suspects.
Predicted Genre: thriller
-----
Description:
Two people from different backgrounds meet and gradually fall in love.
Predicted Genre: drama
-----
Description:
A family goes on an exciting journey across the country and faces many unexpected adventures.
Predicted Genre: documentary
-----
```

The sample predictions demonstrate that the trained NLP model can be used to classify previously unseen movie descriptions.

8. Limitations

The project has the following limitations:

1. **Limited number of genres:** Only eight genres were selected from the original 28 categories. Therefore, the current system cannot classify movies into every genre present in the dataset.
2. **Class imbalance:** The selected genres do not contain exactly equal numbers of movie descriptions. This may cause the model to perform better for some genres than others.
3. **Similar genre characteristics:** Some genres have overlapping themes and vocabulary. For example, Action, Horror, and Thriller movies may contain similar words, making them difficult to distinguish.
4. **Limited contextual understanding:** TF-IDF focuses mainly on the importance of words and phrases and does not deeply understand the context or meaning of a sentence.
5. **Classification errors:** The final accuracy of 69.84% indicates that some movie descriptions are incorrectly classified.

6. **Difficulty with unseen vocabulary:** The model may perform poorly when a new description contains words or expressions that were not sufficiently represented in the training data.
7. **Basic preprocessing:** The project uses relatively simple preprocessing techniques. More advanced linguistic processing could potentially improve the results.
8. **Single-label prediction:** The system predicts one genre for each movie, even though a movie may naturally belong to multiple genres.

9. Future Scope

The project can be improved in several ways:

1. **Use a larger and more balanced dataset:** Additional movie descriptions can be used to improve the representation of less frequent genres.
2. **Include all available genres:** The system can be extended to classify all 28 genre categories in the original dataset.
3. **Improve preprocessing:** Advanced techniques such as tokenization, lemmatization, stemming, spelling correction, and handling of named entities can be explored.
4. **Use advanced machine learning models:** Algorithms such as Support Vector Machines, Random Forest, or other classifiers can be compared with the current models.
5. **Use deep learning:** Neural network architectures such as RNN, LSTM, or GRU could be explored for better sequence and contextual understanding.
6. **Use transformer-based models:** Models such as BERT and its variants could be used to capture deeper semantic relationships in movie descriptions.
7. **Multi-label genre classification:** Future versions could allow a movie to be assigned multiple genres instead of only one.
8. **Web application:** The trained model could be deployed through a web application where users enter a movie description and receive a predicted genre.

9. **Movie recommendation integration:** Genre prediction could be combined with a movie recommendation system to recommend similar movies.
10. **Real-time application:** The system could be integrated into movie databases or streaming platforms for automatic genre classification.

10. Conclusion

This project developed an **NLP-based movie genre classification system** that predicts the genre of a movie using its textual description.

The project used a movie dataset containing **108,414 original records and four attributes**. Eight well-represented genres were selected for the classification task, and after data preparation, the final dataset contained **90,496 movie records**.

The movie descriptions were cleaned and preprocessed before being converted into numerical features using **TF-IDF with unigrams and bigrams**. Two machine learning algorithms, **Logistic Regression** and **Multinomial Naive Bayes**, were implemented and compared.

Logistic Regression achieved the best accuracy of **69.84%**, while Multinomial Naive Bayes achieved **63.55%**. Therefore, Logistic Regression was selected as the final model.

The project also demonstrated how the trained model can be used to classify new movie descriptions. The implementation provided practical experience in dataset preparation, text preprocessing, TF-IDF feature extraction, machine learning classification, model evaluation, and prediction.

Overall, the project demonstrates that traditional NLP techniques combined with machine learning can be effectively used to perform movie genre classification, while also showing the limitations of TF-IDF-based approaches when dealing with complex semantic relationships between different genres.